Bayesian Spatiotemporal Modeling of Passenger Trip Assignment in Metro Networks

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April 8, 2026

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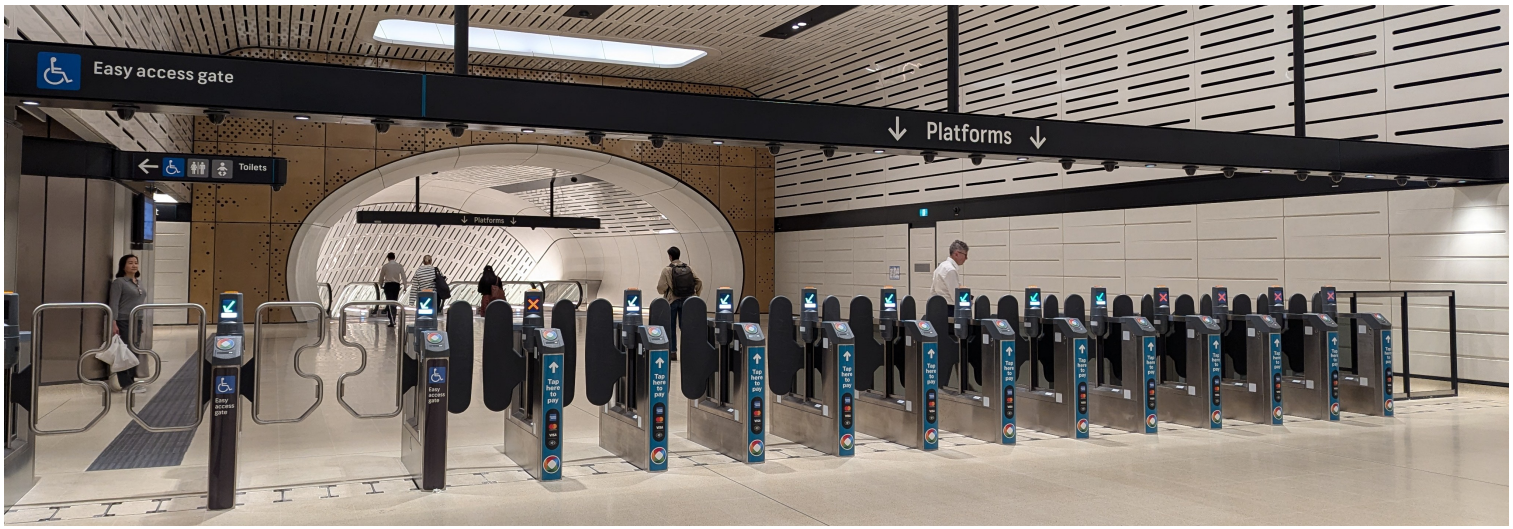
Lijun Sun

McGill University

Chen, Xiaoxu, Alexandra M. Schmidt, Zhenliang Ma, and Lijun Sun. "Bayesian spatiotemporal modeling of passenger trip assignment in metro networks." arXiv preprint arXiv:2507.22403 (2025). (Major Revision at Annals of Applied Statistics)

AFC data in metro systems

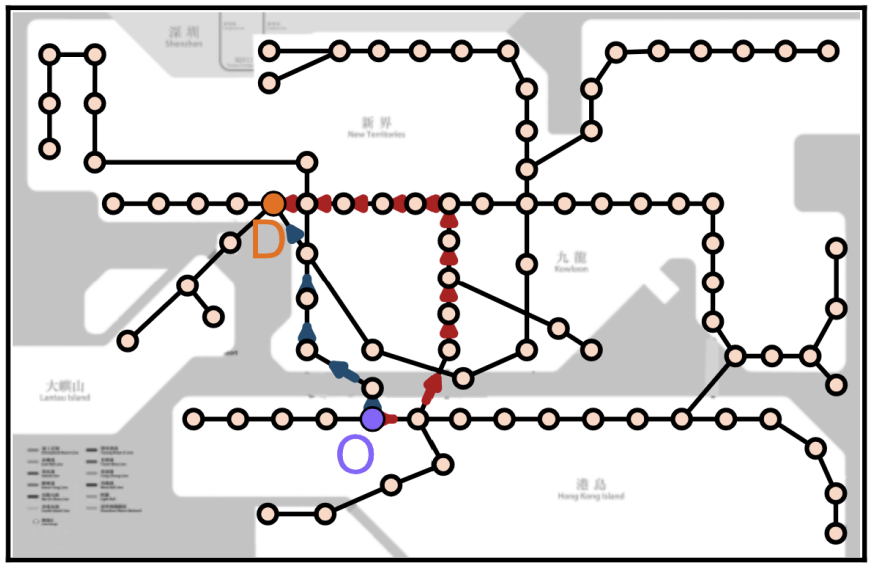
- AFC (automatic fare collection) systems are widely used in metro systems, providing detailed information for each passenger trip:
  - trip ID
  - origin station, destination station
  - tap-in time, tap-out time



Source: https://commons.wikimedia.org/wiki/File:Victoria_Cross_Station_Sydney_13.jpg

What is passenger trip assignment?

- Determining the specific paths passengers take from origin to destination within the metro network.



- The assignment results are essential for optimizing transit service, such as network design, train scheduling, and fare policy.

Research question

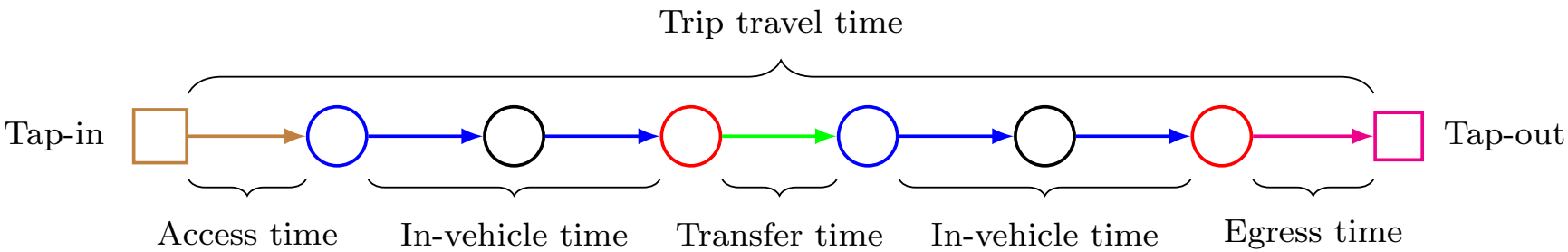
- In metro systems, passenger paths are typically unobservable.
- In this talk, we focus on how to infer passenger paths using AFC data?

Problem formulation

- For any origin-destination (O-D) pair (o, d), there exists a set of feasible paths, $R_{od} = \{r_{od}^k\}_{k=1}^{|R_{od}|}$. When multiple paths exist ($|R_{od}| > 1$), passengers choose a path based on various factors.
- Our observed AFC data consist of a set of trip travel times, $Y = \{y_{odt}^m\}$, where each observation y_{odt}^m is the travel time for trip m between O-D pair (o, d) starting in time interval t.
- Our primary objective is to infer the latent path choice, $z_{odt}^m \in \{1, \ldots, |R_{od}|\}$, for each trip observation.

A model for trip travel time

- To model the observed travel time y_{odt}^m, we decompose it into four components:
  - Access time: from entry tap-in to boarding
  - In-vehicle time: spent on the train
  - Transfer time: the duration between alighting and boarding connecting trains
  - Egress time: from alighting to exit tap-out



A model for trip travel time

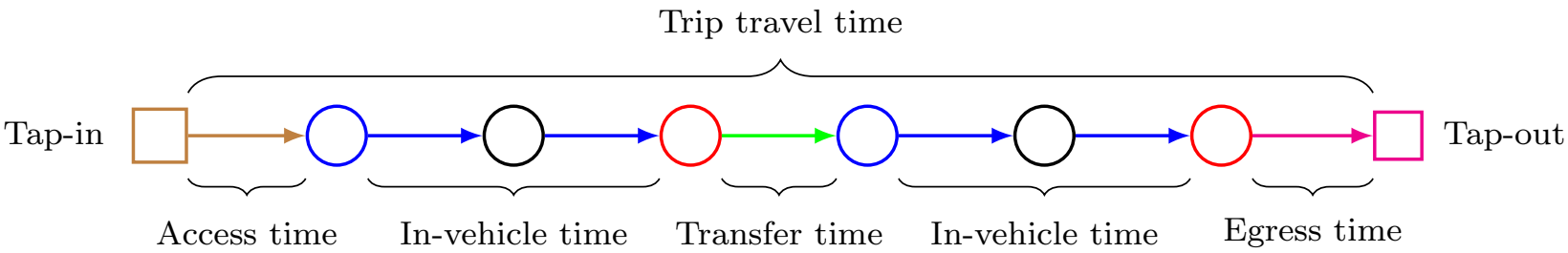
- Represent the costs associated with these components as vectors: access costs $\boldsymbol{a} \in \mathbb{R}^l$, in-vehicle costs $\boldsymbol{h} \in \mathbb{R}^l$, transfer costs $\boldsymbol{u} \in \mathbb{R}^s$, and egress costs $\boldsymbol{e} \in \mathbb{R}^n$.
- Combine them into a single network cost vector, $\boldsymbol{x} = (\boldsymbol{a}^\top, \boldsymbol{h}^\top, \boldsymbol{u}^\top, \boldsymbol{e}^\top)^\top \in \mathbb{R}^c$ where $c = 2l + n + s$.
- As network conditions are dynamic, these costs vary over time. We model a sequence of time-varying cost vectors, $\{\boldsymbol{x}_t\}_{t=1}^T$.

A model for trip travel time

- For a given path choice $z_{odt}^m = k$, the observed travel time y_{odt}^m can be expressed as a linear function of the network costs at time t:

$$y_{odt}^m \mid (\boldsymbol{x}_t, z_{odt}^m = k) = \boldsymbol{A}_{od}^k \boldsymbol{x}_t + \varepsilon_{odt}^m,$$

where $\boldsymbol{A}_{od}^k$ is a binary routing matrix for path r_{od}^k and ε_{odt}^m is a zero-mean error term.



- It is a challenging inverse problem. Only the total travel time y_{odt}^m is observed. The two key variables (the latent path choice z_{odt}^m and the dynamic network costs $\boldsymbol{x}_t$) are unobserved and must be inferred simultaneously from the data.

A heteroscedastic model for trip travel times

For the above linear model: $y_{odt}^m \mid (\boldsymbol{x}_t, z_{odt}^m = k) = \boldsymbol{A}_{od}^k \boldsymbol{x}_t + \varepsilon_{odt}^m$, how to model the error term ε_{odt}^m?

- A simple homoscedastic assumption for the error term ($\varepsilon_{odt}^m \sim \mathcal{N}(0, \sigma^2)$) is unrealistic, as longer and more complex paths are inherently more variable.
- To capture this heterogeneity in a parsimonious way, we model the variance of the error term as being proportional to the path's expected cost components. Specifically, we define the error as:

$$\varepsilon_{odt}^m \mid \boldsymbol{x}_t, \boldsymbol{\sigma}, z_{odt}^m = k \sim \mathcal{N}\left(0, \boldsymbol{A}_{od}^k (\boldsymbol{x}_t * \boldsymbol{\sigma})^2\right),$$

where $*$ is the element-wise Hadamard product and $\boldsymbol{\sigma}$ is a vector of coefficients of variation.

A heteroscedastic model for trip travel times

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- For simplicity, we assume a common coefficient for each cost type: $\boldsymbol{\sigma} = (\sigma_a \mathbf{1}_l^\top, \sigma_h \mathbf{1}_l^\top, \sigma_u \mathbf{1}_s^\top, \sigma_e \mathbf{1}_n^\top)^\top$. Combining these elements gives the full observation equation:

$$y_{odt}^m \mid \boldsymbol{x}_t, \boldsymbol{\sigma}, z_{odt}^m = k \sim \mathcal{N}\left(\boldsymbol{A}_{od}^k \boldsymbol{x}_t, \boldsymbol{A}_{od}^k (\boldsymbol{x}_t * \boldsymbol{\sigma})^2\right).$$

A random walk for dynamic network costs

- Model dynamics of the network cost vector $\boldsymbol{x}_t$ using a random walk:

$$\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1} \sim \mathcal{N}\left(\boldsymbol{x}_{t-1}, \mathsf{diag}(\boldsymbol{\tau}^2)\right), \quad t = 2, \ldots, T,$$
$$\boldsymbol{x}_1 \sim \mathcal{N}(\boldsymbol{m}_0, \boldsymbol{P}_0).$$

Path choice model: MNL with spatiotemporal coefficients

- MNL choice model is widely used for modeling passenger path choices in metro systems.
- Previous studies often assume that all passengers have the same utility parameters regardless of space and time. However, passenger preferences should not be static but vary across space and time.
- The utility V_{odt}^{k} of a path r_{od}^{k} is defined as a linear combination of the in-vehicle and transfer costs it comprises.

Path choice model: MNL with spatiotemporal coefficients

- We allow the coefficients to vary across origin o and interval t:

$$V_{odt}^{k} = \theta_{ot} \sum_{a \in v_{od}^{k}} x_{t}^{a} + \beta_{ot} \sum_{a \in u_{od}^{k}} x_{t}^{a},$$

  where θ_{ot} and β_{ot} are the spatiotemporally varying sensitivities to in-vehicle and transfer time, respectively.

- The probability of choosing path k is then given by the standard logit formula:

$$p(z_{odt}^{m} = k \mid \boldsymbol{x}_t, \theta_{ot}, \beta_{ot}) = \frac{\exp(V_{odt}^{k})}{\sum_{k'=1}^{|R_{od}|} \exp(V_{odt}^{k'})}.$$

Path choice model: MNL with spatiotemporal coefficients

How to model spatiotemporally varying coefficients $\boldsymbol{\Theta} = \{\theta_{ot}\} \in \mathbb{R}^{n \times T}$ and $\boldsymbol{\Phi} = \{\beta_{ot}\} \in \mathbb{R}^{n \times T}$?

- A natural method is to use GP as follows:

$$\operatorname{vec}(\boldsymbol{\Theta}) \sim \mathcal{N}(\mathbf{0}_{nT}, \boldsymbol{K}_S \otimes \boldsymbol{K}_T),$$
$$\operatorname{vec}(\boldsymbol{\Phi}) \sim \mathcal{N}(\mathbf{0}_{nT}, \boldsymbol{K}_S \otimes \boldsymbol{K}_T),$$

  where $\boldsymbol{K}_S \in \mathbb{R}^{n \times n}$ and $\boldsymbol{K}_T \in \mathbb{R}^{T \times T}$ are the spatial and temporal covariance matrices, respectively.

- However, the dimension of $\operatorname{vec}(\boldsymbol{\Theta})$ and $\operatorname{vec}(\boldsymbol{\Phi})$ is nT, which presents a significant computational challenge to achieve convergence due to the large parameter space.

Path choice model: MNL with spatiotemporal coefficients

How to model spatiotemporally varying coefficients $\boldsymbol{\Theta} = \{\theta_{ot}\} \in \mathbb{R}^{n \times T}$ and $\boldsymbol{\Phi} = \{\beta_{ot}\} \in \mathbb{R}^{n \times T}$?

- To address the high-dimensionality challenge, we represent them as a single parameter tensor $\boldsymbol{\mathcal{F}} \in \mathbb{R}^{2 \times n \times T}$ and use tensor decomposition:

$$\boldsymbol{\mathcal{F}} = \boldsymbol{\mathcal{Q}} + \sum_{r=1}^{R} \boldsymbol{u}_r \circ \boldsymbol{v}_r \circ \boldsymbol{w}_r.$$

- $\boldsymbol{u}_r \in \mathbb{R}^2$, $\boldsymbol{v}_r \in \mathbb{R}^n$, and $\boldsymbol{w}_r \in \mathbb{R}^T$ are the latent factor vectors corresponding to choice (parameter) type, station, and time.
- The CP decomposition significantly reduces the number of parameters from $2nT$ to $(2 + n + T)R + 2$.

Path choice model: MNL with spatiotemporal coefficients

How to model spatiotemporally varying coefficients $\boldsymbol{\Theta} = \{\theta_{ot}\} \in \mathbb{R}^{n \times T}$ and $\boldsymbol{\Phi} = \{\beta_{ot}\} \in \mathbb{R}^{n \times T}$?

- We place GP priors on the spatial and temporal factor vectors:

$$\begin{aligned} \boldsymbol{v}_r &\sim \mathcal{N}(\mathbf{0}_n, \boldsymbol{K}_S), \quad r = 1, \ldots, R, \\ \boldsymbol{w}_r &\sim \mathcal{N}(\mathbf{0}_T, \boldsymbol{K}_T), \quad r = 1, \ldots, R. \end{aligned}$$

- For factor matrix $\boldsymbol{U}$, we assume the prior $\boldsymbol{u}_r \sim \mathcal{N}(\mathbf{0}_2, \boldsymbol{K}_U)$. Further, we impose an inverse Wishart prior on $\boldsymbol{K}_U$:

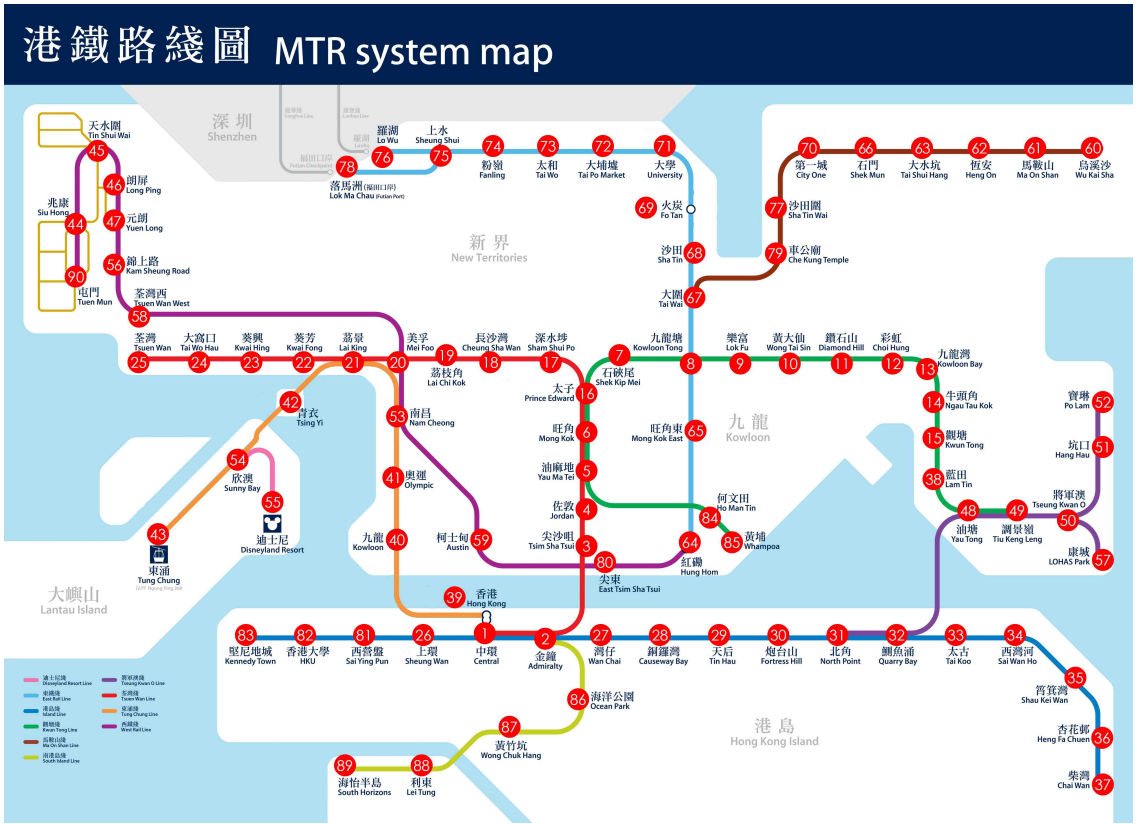
$$\boldsymbol{K}_U \sim \mathcal{W}^{-1}(\boldsymbol{\Omega}_0, \nu_0).$$

Markov chain Monte Carlo (MCMC)

- From the joint posterior distribution, we factorize the problem into four manageable blocks for our Gibbs sampler:
  - Draw dynamic network costs, $\{\boldsymbol{x}_t\}$ with FFBS algorithm.
  - Draw coefficients of variation, $\{\sigma_k\}$ with slice sampling.
  - Draw the parameters $\boldsymbol{V}, \boldsymbol{W}$ with Elliptical slice sampling, $\boldsymbol{U}$ from its inverse Wishart posterior, and baseline effects q_1, q_2 with slice sampling, and get spatiotemporal choice parameters.
  - Draw the latent path choice outcomes $\boldsymbol{Z}$ from categorical posterior.

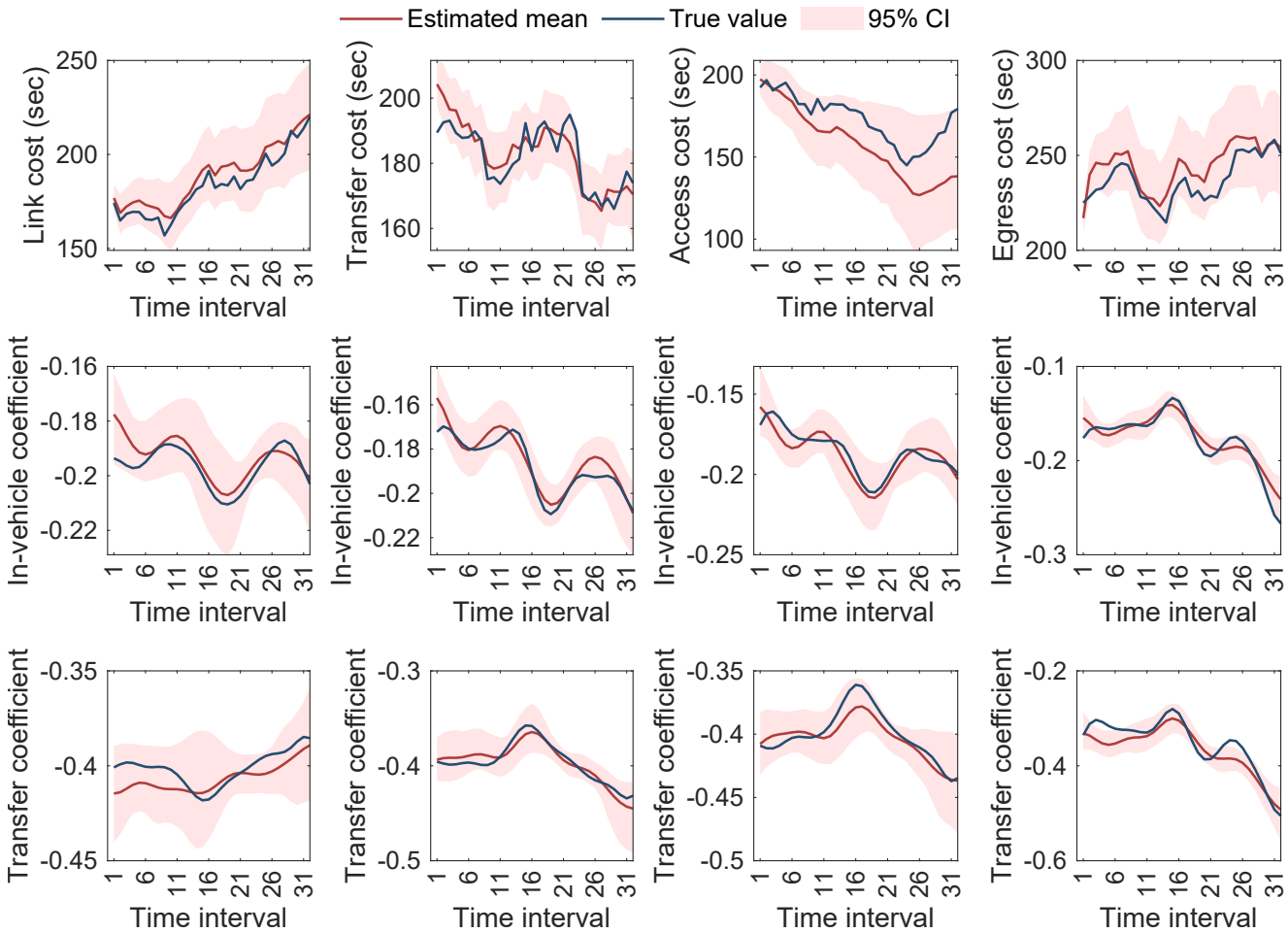
Simulation study

- We use the network topology of the Hong Kong MTR system. We generate a synthetic dataset by a set of ground-truth parameters. Validate if our inference method can recover the parameters.



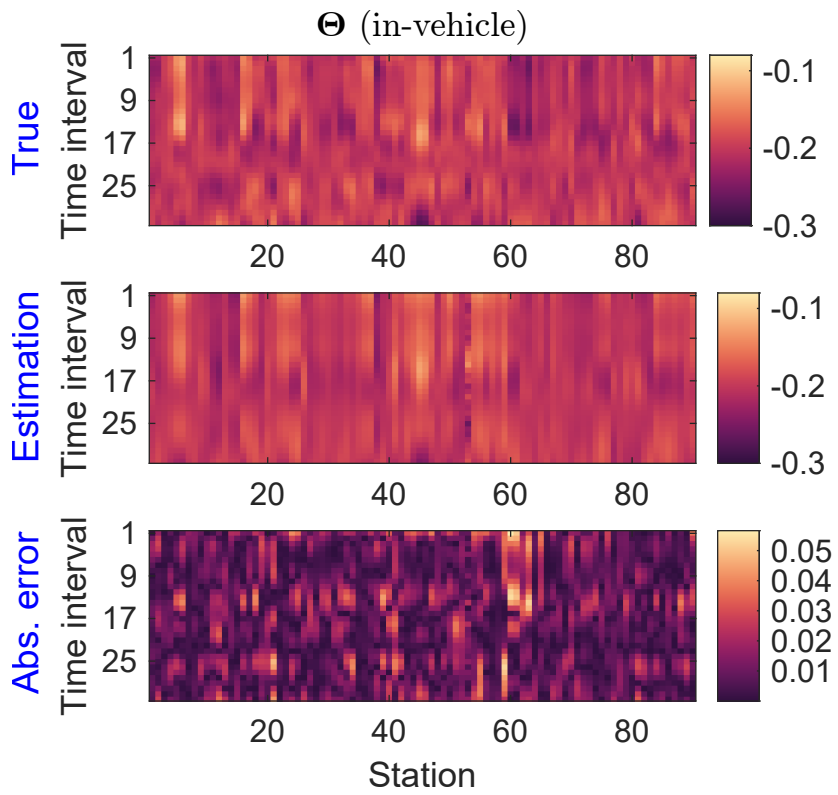
Simulation study

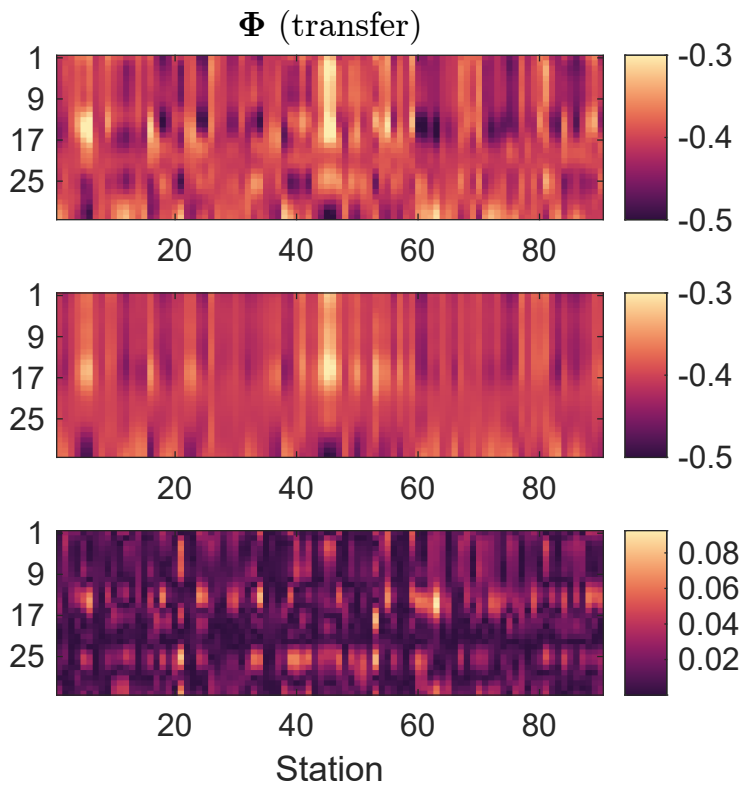
- Recovery of simulation parameters.



Simulation study

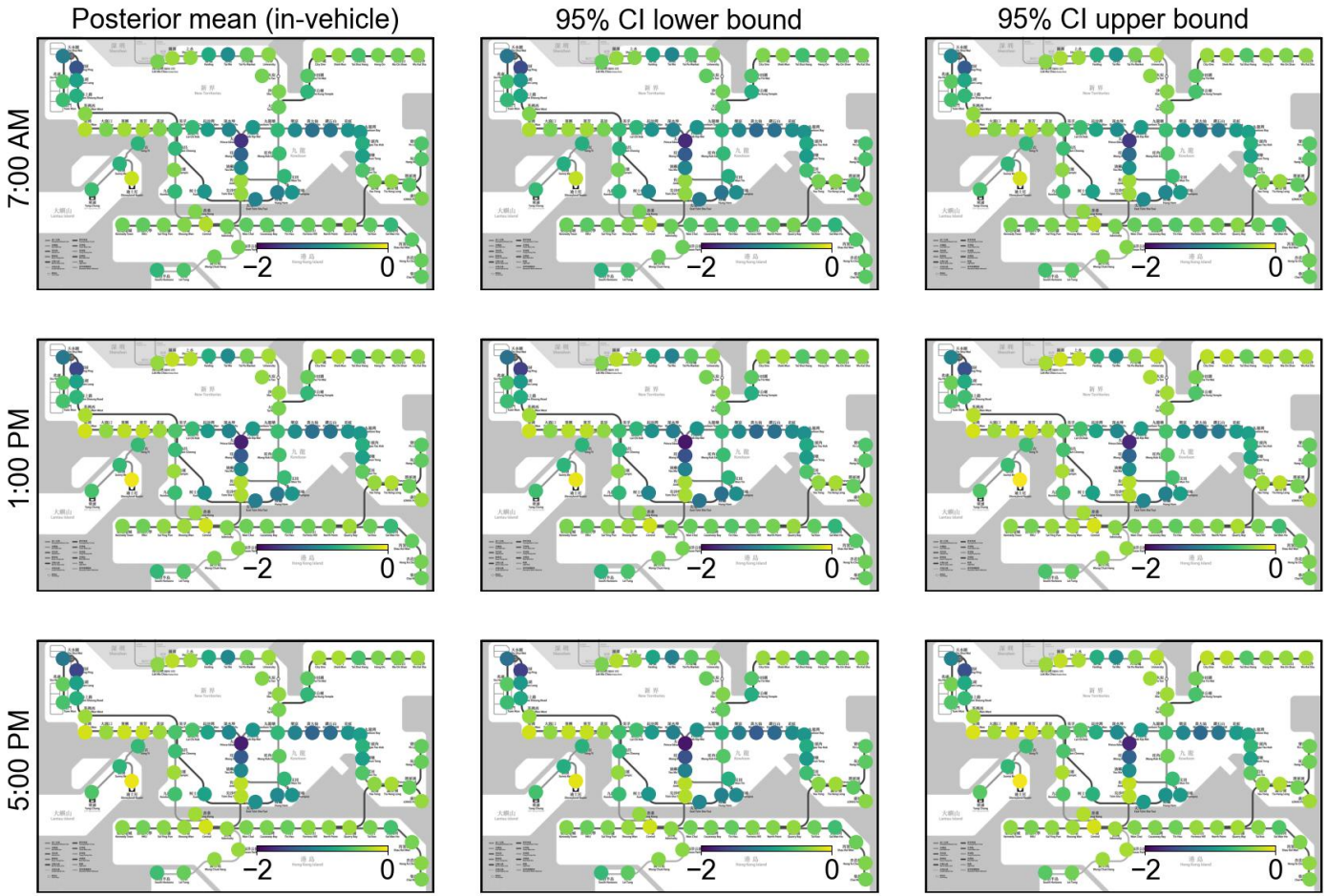
- Recovery of spatiotemporal choice coefficients.





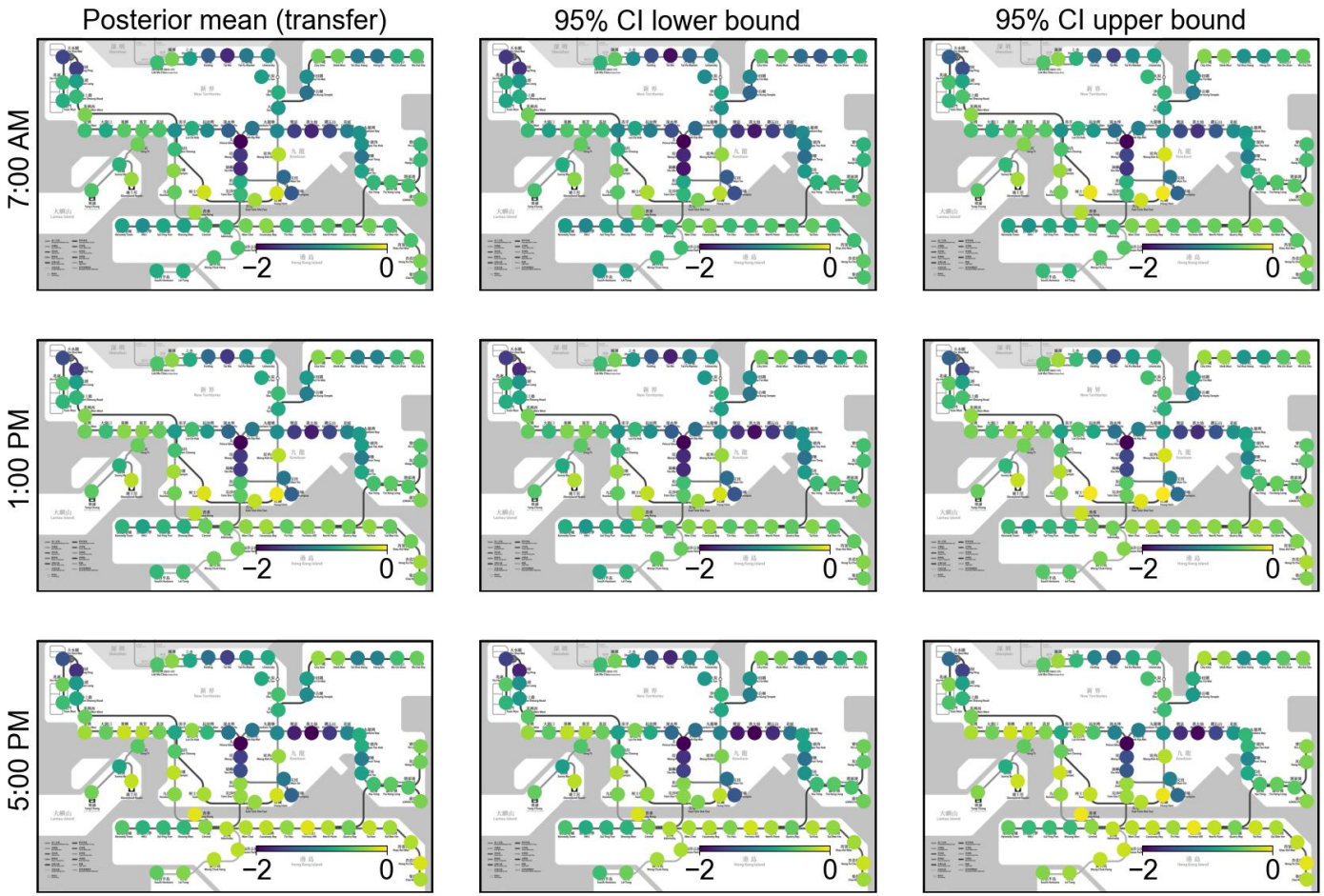
Application to real-world metro data

- Use AFC data from Hong Kong (> 4 million records one weekday)
- The estimated posterior mean and 95% CIs of spatiotemporal $\boldsymbol{\Theta}$.



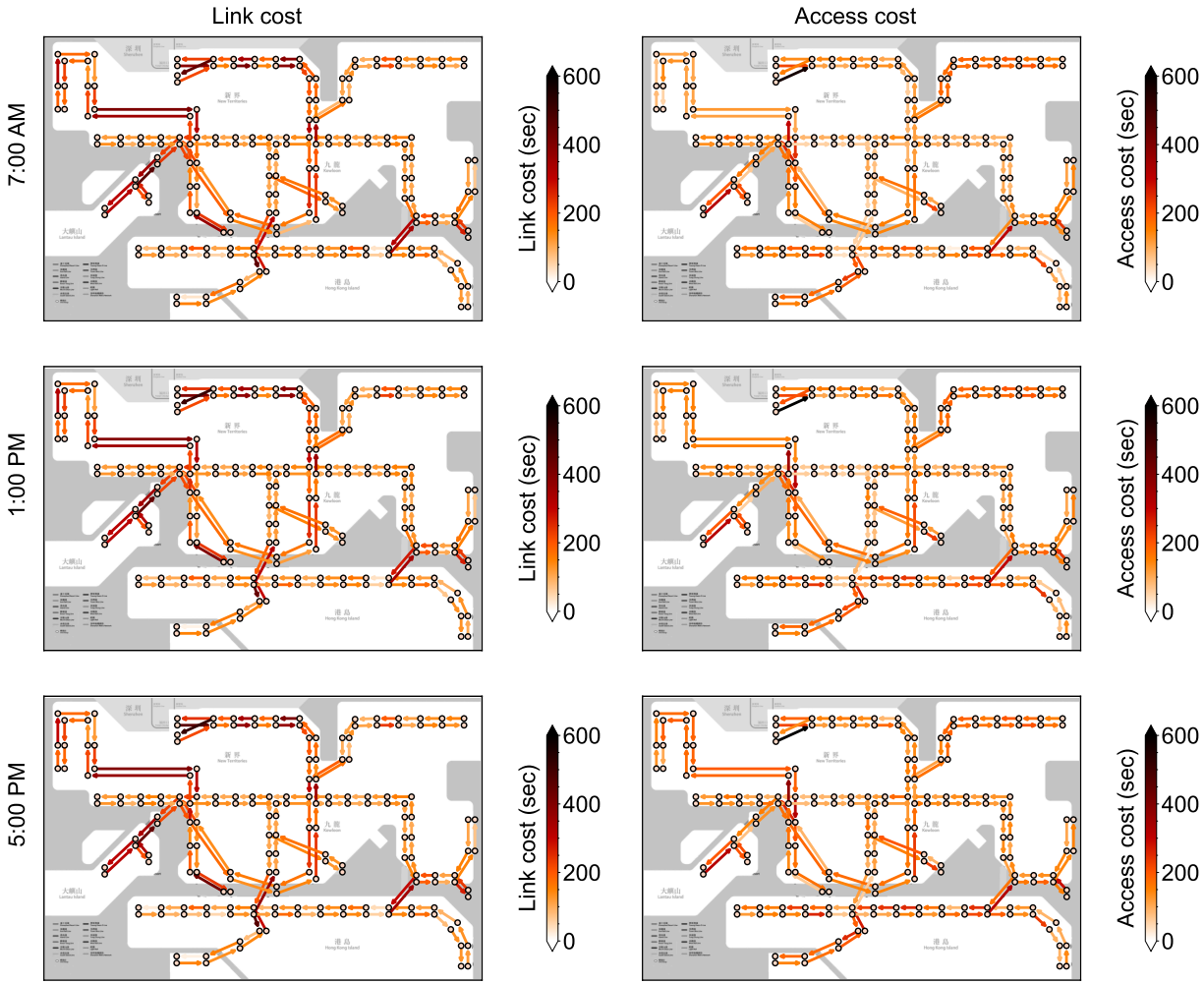
Application to real-world metro data

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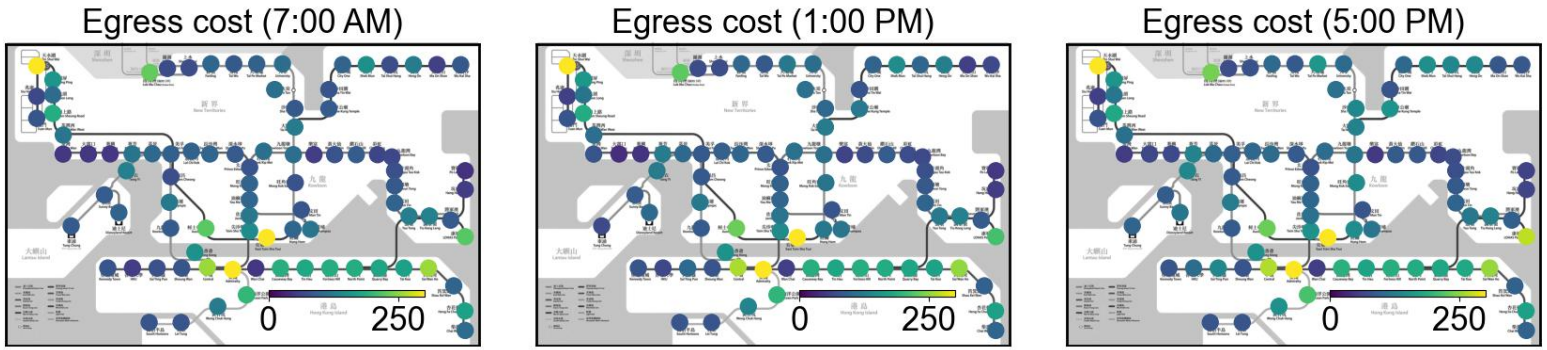
Application to real-world metro data

- Estimated link and access costs in the metro networks.

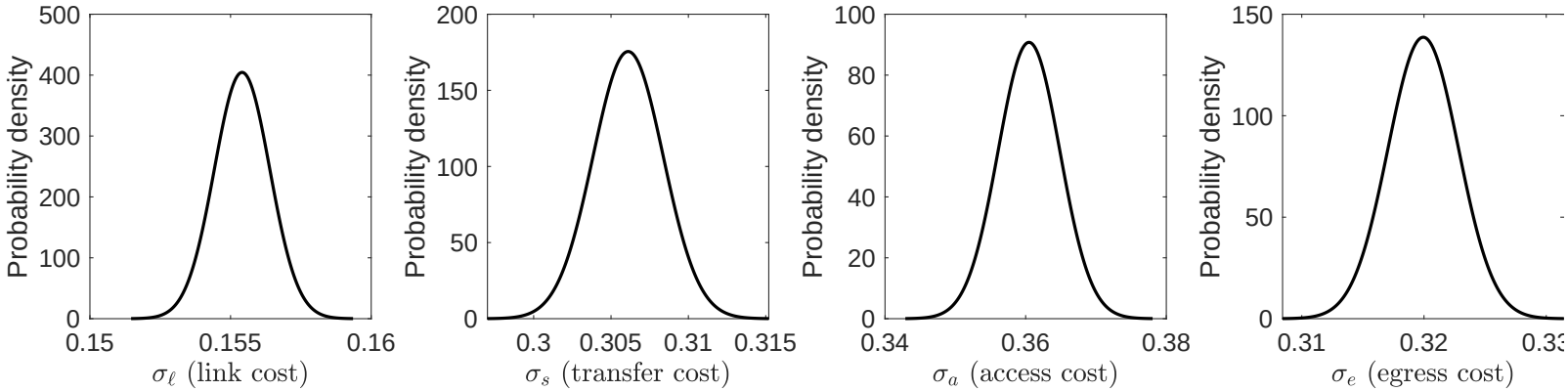


Application to real-world metro data

- Estimated egress costs in the metro networks.

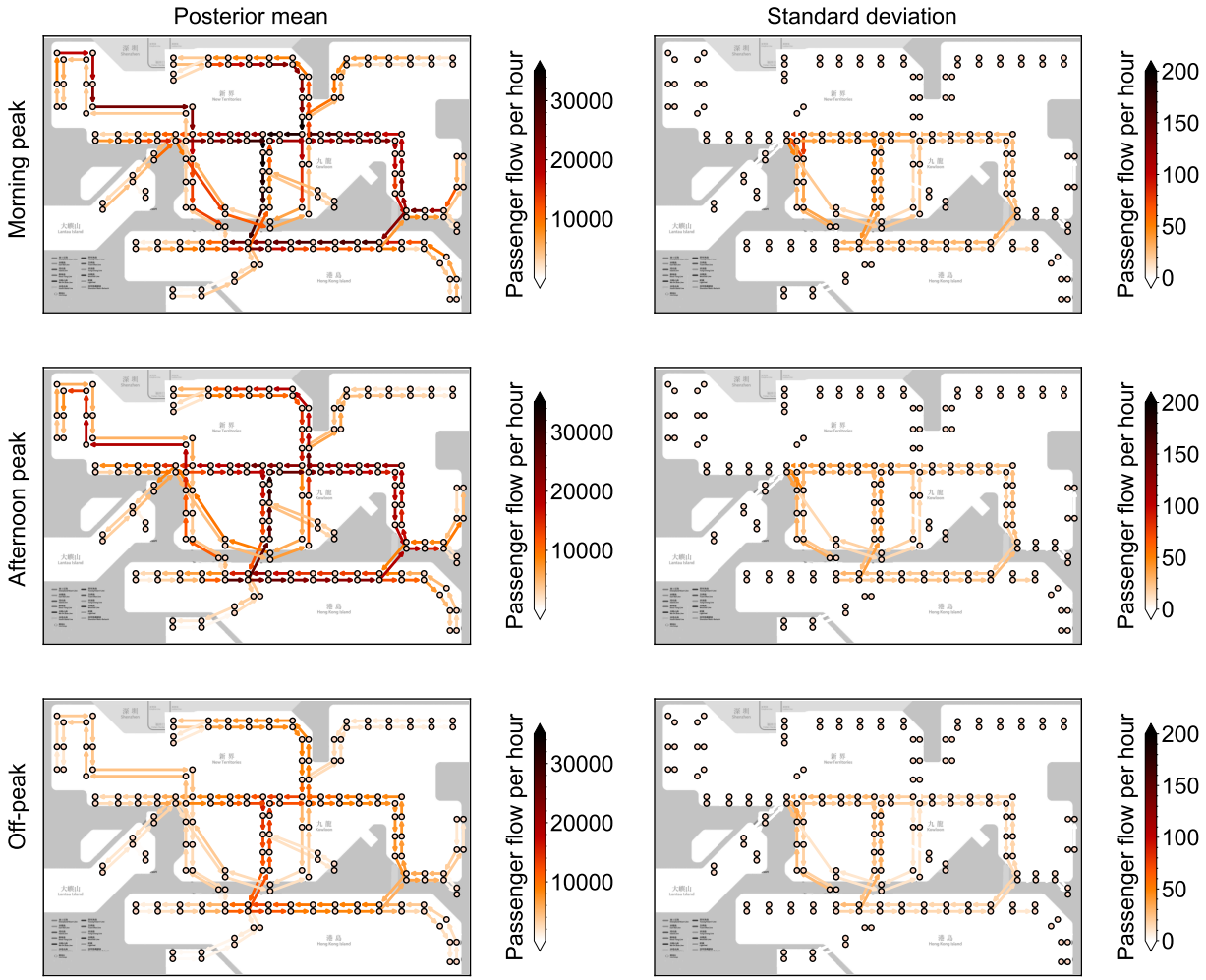


- The estimated coefficient of variation parameters for network costs.



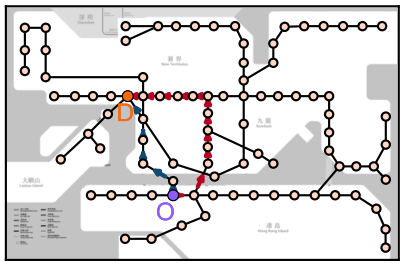
Application to real-world metro data

- The posterior mean and std of passenger flow assignment.

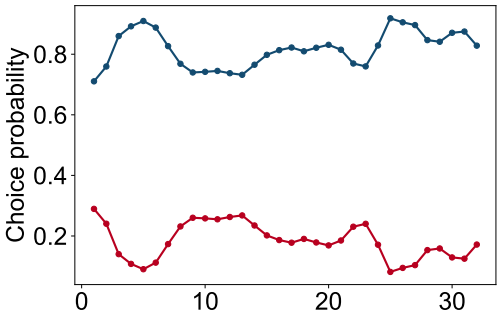


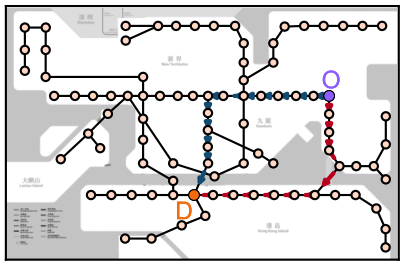
Application to real-world metro data

- Temporal variation of path choice probabilities for three O-D pairs.

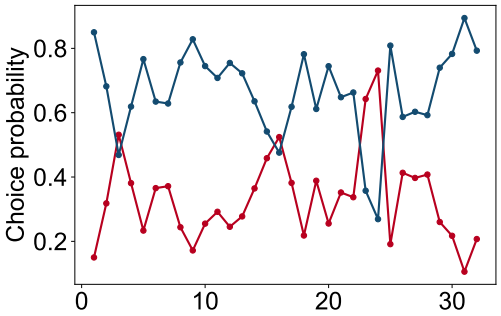


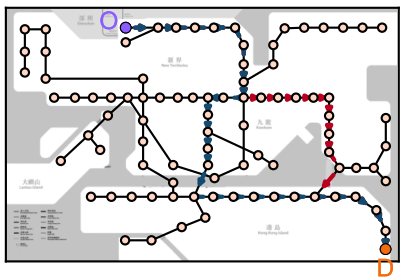
O-D pair (A)



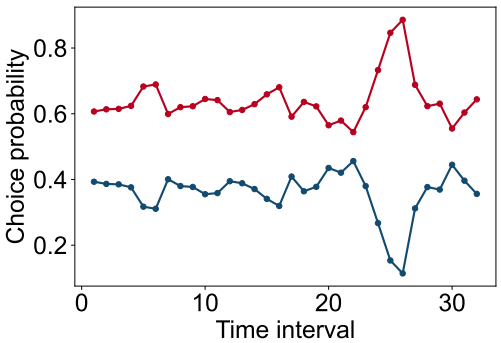


O-D pair (B)





O-D pair (C)



Summary

The main contributions of this work are:

- Propose a Bayesian framework that provides a dynamic and accurate representation of passenger behavior by jointly estimating time-varying network costs and path choices.
- Propose an MNL choice model with spatiotemporally varying coefficients that captures the evolving nature of passenger preferences for in-vehicle and transfer costs.
- Develop an efficient MCMC-based inference algorithm, using an FFBS sampler, ESS, and slice sampling.

Thank you!

Questions?